QUESTION NO. 25

COBOL PROGRAM FOR BANK ACCOUNT INTEREST CALCULATION

1. Aim

To develop a COBOL program that calculates the monthly interest for multiple bank accounts using each account's balance and annual interest rate. The program displays the interest for each account and the total interest for all accounts.

2. Algorithm

1. Start the program.
2. Read the number of bank accounts to be processed.
3. Initialize the total interest to zero.
4. Read the account ID, balance, and annual interest rate for each account.
5. Calculate monthly interest using Balance × Rate ÷ 1200.
6. Add the calculated interest to the total interest.
7. Display the account ID and its monthly interest.
8. Repeat the process until all accounts are processed.
9. Display the total interest for all accounts.
10. Stop the program.

3. COBOL Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. INTEREST.

DATA DIVISION.
WORKING-STORAGE SECTION.

01 N PIC 9(5).
01 I PIC 9(5).
01 ACCOUNT-ID PIC X(10).
01 BALANCE PIC 9(7).
01 RATE PIC 99.
01 INTEREST PIC 9(7)V99.
01 TOTAL PIC 9(9)V99 VALUE 0.
01 OUT-INTEREST PIC Z(6)9.99.
01 OUT-TOTAL PIC Z(8)9.99.

PROCEDURE DIVISION.
    ACCEPT N

    PERFORM VARYING I FROM 1 BY 1
        UNTIL I > N

        ACCEPT ACCOUNT-ID
        ACCEPT BALANCE
        ACCEPT RATE

        COMPUTE INTEREST ROUNDED =
            BALANCE * RATE / 1200

        ADD INTEREST TO TOTAL

        MOVE INTEREST TO OUT-INTEREST

        DISPLAY FUNCTION TRIM(ACCOUNT-ID)
            " " FUNCTION TRIM(OUT-INTEREST)

    END-PERFORM

    MOVE TOTAL TO OUT-TOTAL
    DISPLAY "Total interest: "
        FUNCTION TRIM(OUT-TOTAL)

    STOP RUN.
```

4. Workflow

START
Enter the number of accounts
Read account ID, balance and annual interest rate
Calculate monthly interest = Balance × Rate ÷ 1200
Add interest to the running total
Display account ID and monthly interest
Are all accounts processed?
NO → Read the next account | YES → Display total interest
STOP

5. Sample Input

```
2
A100
10000
6
A200
20000
3
```

6. Expected Output

```
A100 50.00
A200 50.00
Total interest: 100.00
```

7. Result

The COBOL program calculates the monthly interest for each bank account, displays the individual interest amounts, and displays the total interest for all accounts.